Bitcoin Market Risk Analysis

April 13, 2026

Executive Summary

Bitcoin remains one of the most volatile liquid assets in global markets, which makes it a strong stress case for market risk methodology. This study builds a five-year hourly risk engine from January 2021 through the current date using BTC/USDT spot data from Binance as a USD proxy. The design goal is not only to report a single Value at Risk number, but to show why a heavy-tailed, simulation-based, and neural-network-enhanced workflow is justified for crypto assets whose distributions are non-Gaussian, clustered in volatility, and punctuated by jumps. At the same time, the empirical conclusions must be read with explicit caution: the benchmark uses a stablecoin proxy rather than a true fiat BTC/USD venue, the neural models are only partially validated out of sample, and the reported VaR figures are market-risk estimates rather than liquidity-adjusted liquidation costs.

Data Engineering

The dataset was built from hourly Binance BTC/USDT spot candles beginning on 2021-01-01T00:00:00+00:00 and ending on 2026-04-13T06:00:00+00:00. BTC/USDT is used as a highly liquid USD proxy because it offers uninterrupted hourly depth across the full study horizon. The cleaning process enforces a continuous hourly timestamp index, repairs small data outages, removes duplicates, and flags flash-crash artefacts only when they look like data-feed anomalies rather than genuine market moves.

That choice is operationally convenient but not theoretically neutral. BTC/USDT embeds not only Bitcoin market risk but also residual Tether credit and peg risk. During stress episodes such as the May 2022 stablecoin dislocation, temporary USDT/USD deviations can contaminate the measured return distribution and therefore bias both volatility and tail-loss estimates. A production-grade extension should therefore either migrate to true fiat pairs such as Coinbase or Kraken BTC/USD, or explicitly deflate the BTC/USDT series by a synchronized USDT/USD basis series before computing returns.

Original rows	Clean rows	Missing hours	Duplicates removed	Flash-crash repairs	Start	End
46265	46279	14	0	0	2021-01-01 00:00 UTC	2026-04-13 06:00 UTC

Methodology

The pipeline begins with market data engineering. Hourly candles are fetched directly from the exchange API, then reindexed to a continuous hourly clock to eliminate gaps in the time axis.

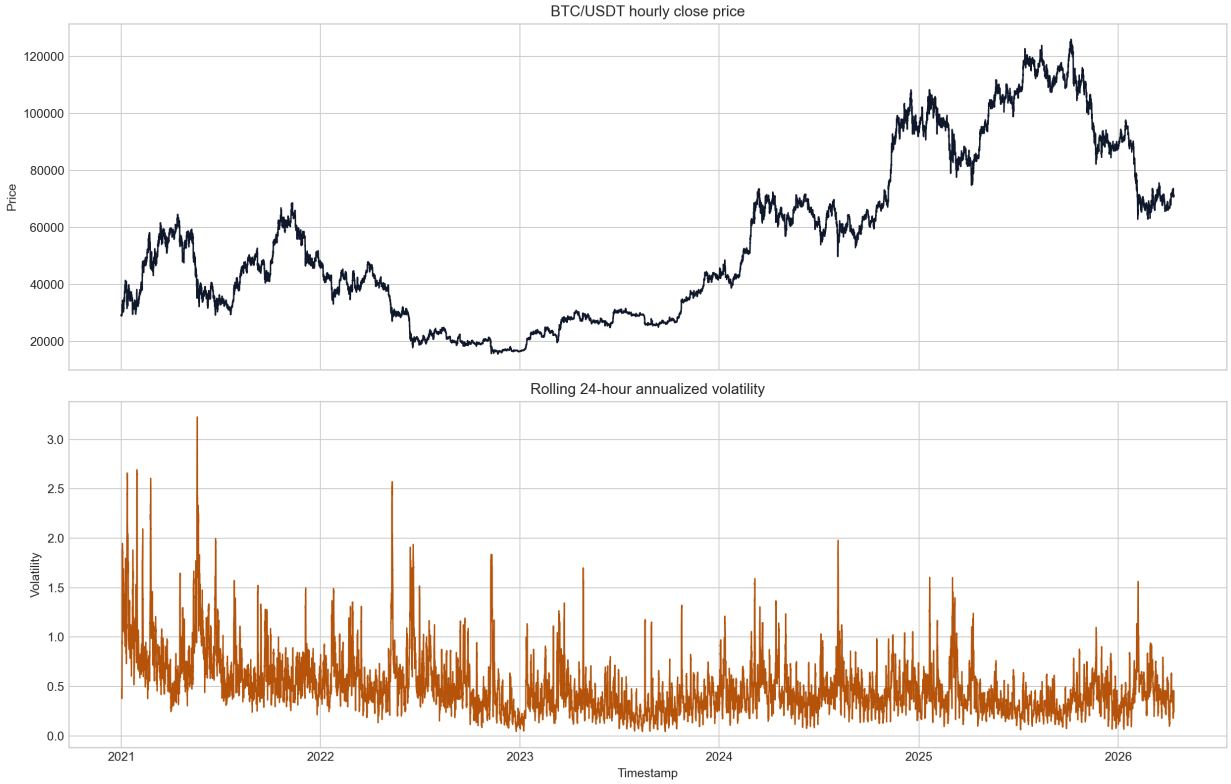

Figure 1: Price and volatility

Missing bars are interpolated conservatively, duplicates are removed, and single-bar flash-crash artefacts are repaired only when the move is immediately reversed and the net two-hour move is small. This preserves genuine market shocks while reducing obvious feed noise. Log returns are then calculated, followed by rolling 24-hour and 168-hour volatility, RSI, MACD, and a rolling volume z-score. These engineered features feed the downstream conditional risk models.

The statistical section tests whether simple Gaussian assumptions are defensible. Jarque-Bera quantifies departures from normality, skewness and kurtosis measure asymmetry and tail thickness, and the Augmented Dickey-Fuller test evaluates whether the return series is stationary enough to support modeling. In practice, Bitcoin hourly returns typically fail normality decisively while remaining stationary in mean, which is the exact combination that motivates Student-t VaR, Monte Carlo simulation with jumps, and neural network volatility forecasting.

Four model families are implemented. Method A is a parametric Student-t VaR, calibrated on a rolling return window selected by validation, to capture excess kurtosis in a compact distributional form. Method B is a 10,000-path jump-diffusion Monte Carlo engine that retains Gaussian diffusion but adds empirically calibrated jump intensity and jump magnitude to reflect crypto discontinuities. Method C is a PyTorch LSTM with a Gaussian output head that predicts next-hour conditional mean and volatility from rolling features and then maps the predictive distribution into VaR and CVaR. Method D is a PyTorch variational autoencoder trained on rolling 24-hour return windows; it learns a latent representation of return dynamics and generates synthetic paths by sampling from latent space.

The final model settings are selected on the pre-backtest training history only, so the comparison remains out of sample. Student-t and jump diffusion are tuned on rolling calibration windows, the LSTM uses the best look-back window from the validation sweep, and the VAE uses the best latent size before any backtest or stress episode is evaluated.

Method	Model	Primary role	Main trade-off
A	Student-t VaR	Parametric heavy-tail VaR using a recent rolling return window.	Transparent and fast, but only weakly conditional on market state.
B	Jump-diffusion Monte Carlo	Scenario engine that combines diffusion risk with discontinuous crypto-style jumps.	More realistic under stress, but sensitive to jump calibration choices.
C	LSTM Gaussian head	Feature-conditioned forecast of next-hour mean and volatility from engineered signals.	Most adaptive model in the stack, but harder to explain and validate.
D	VAE latent sampling	Generative model that learns return-window structure and produces synthetic tail scenarios.	Useful for scenario enrichment, but less direct for one-step production VaR.

Tuned Model Settings

Model	Design	Best parameters	Selection metric	Validation score
Student-t VaR	Parametric heavy-tail baseline	90-day rolling window; df estimated from sample kurtosis	Mean pinball loss	0.0003
Jump-diffusion Monte Carlo	Diffusion plus calibrated jump scenario engine	90-day rolling window; jump threshold 2.5 sigma	Mean pinball loss	0.0003
LSTM cond. VaR	Feature-conditioned recurrent forecaster with Gaussian head	48-hour look-back; 2 layers; 48 hidden units; dropout 0.10	Best validation NLL	-4.7427
VAE latent VaR	Latent generative return model on rolling windows	24-hour window; latent dimension 8	Best validation ELBO	0.5178

The neural models are evaluated with a chronological rather than random split, which is appropriate for financial time series. The last year of the sample is reserved for backtesting, while the earlier history is used for model fitting and the final 20% of the pre-backtest training window is held out for validation. This reduces look-ahead bias, but it is not equivalent to a full rolling-origin cross-validation design across multiple market regimes. Accordingly, strong LSTM performance in the current backtest should be interpreted as encouraging rather than final.

The jump-diffusion engine also requires a qualification. Its jump intensity and jump-size parameters are re-estimated from recent rolling history, which makes the model more adaptive than a single full-sample fit, but the parameters are still treated as piecewise static within each calibration window. In a market whose microstructure has changed from retail-led 2021 momentum to ETF- and institution-influenced 2024–2026 flows, that stationarity assumption can become materially inaccurate.

The downside tail metrics used in the report follow the standard left-tail definitions:

$$CVaR_\alpha = \frac{1}{1-\alpha}\int_\alpha^1 VaR_u \, du$$

For unconditional backtesting, the Kupiec proportion-of-failures statistic is:

$$LR_{POF} = -2\ln\left(\frac{(1-\alpha)^{T-x}\alpha^x}{(1-\hat{p})^{T-x}\hat{p}^x}\right), \qquad \hat{p} = \frac{x}{T}$$

where T is the number of forecasts, x is the number of VaR violations, and $\hat{p}$ is the observed violation rate.

Statistical Diagnostics

Obs.	Mean	Std.	Skew	Kurtosis	JB stat	JB p-val	ADF stat
46278	0	0.0064	-0.2066	18.9551	491079.1521	<0.0001	-30.2084

ADF p-val	ADF 1% crit	ADF 5% crit	ADF 10% crit	Min	Max	P1	P99
<0.0001	-3.4305	-2.8616	-2.5668	-0.0938	0.1161	-0.0193	0.0186

Model Results

The engineered dataset spans 46,279 hourly observations after cleaning, with 14 missing hours filled and 0 flash-crash artefacts repaired. Over the full sample, Bitcoin produced a cumulative price change of 144.96%, while hourly returns ranged from -9.38% to 11.61%. The Jarque-Bera statistic is 491079.15 with p-value 0, which rejects normality by a wide margin. Skewness is -0.207 and kurtosis is 18.955, confirming an asymmetric and leptokurtic distribution. At the same time, the ADF statistic of -30.208 with p-value 0 supports stationarity of hourly returns, so the data are suitable for conditional modeling.

Across the current forecast set, the one-hour VaR estimates differ meaningfully by methodology. Student-t VaR is parsimonious and directly responsive to heavy tails, but it assumes identically distributed shocks over the calibration window. The Monte Carlo model is more flexible because it can separate diffusion from jumps and also extends naturally to multi-period forecasts. The

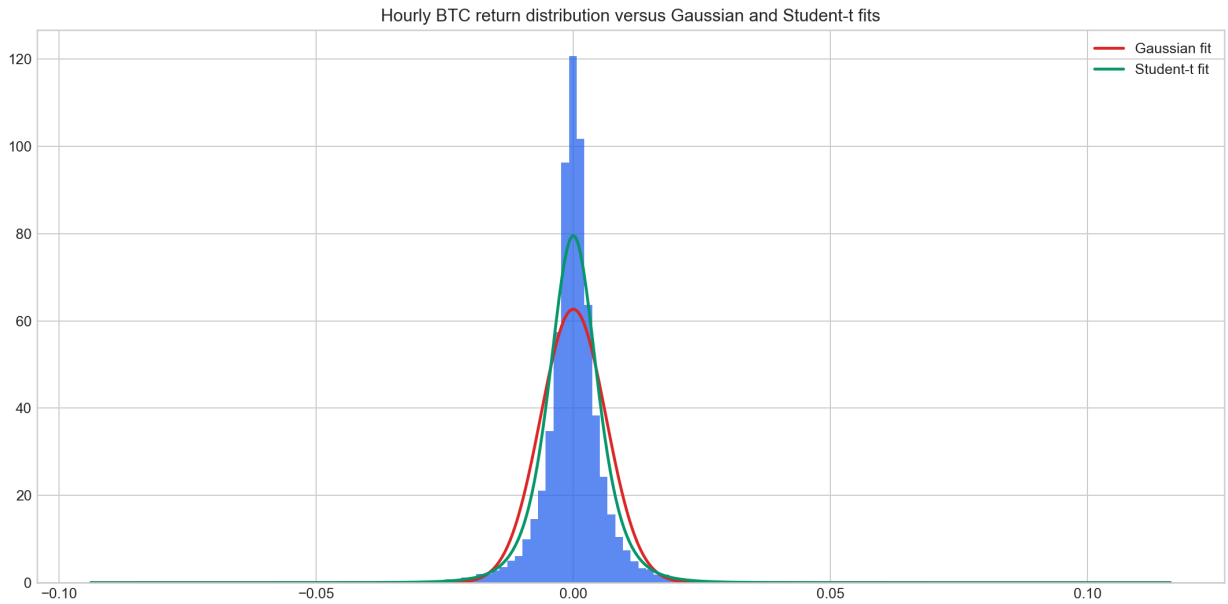

Figure 2: Return distribution

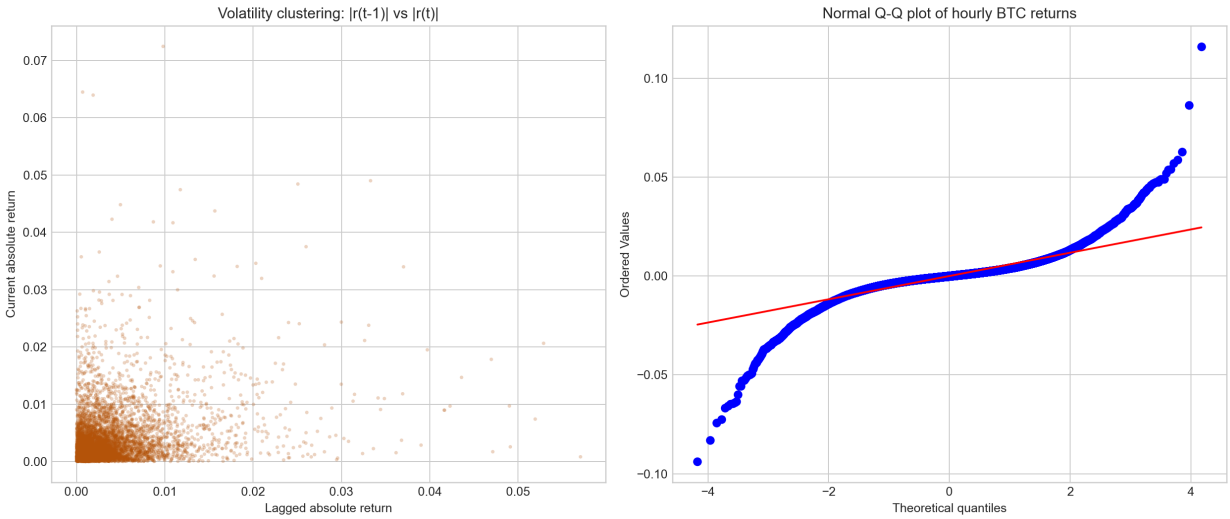

Figure 3: Distribution diagnostics

LSTM reacts to state variables such as recent volatility, RSI, MACD, and abnormal volume, which makes it the most explicitly conditional model in the stack. The VAE is different again: it does not forecast a point volatility path but instead learns a latent manifold of plausible return windows and samples from that manifold to infer risk.

That ranking should not be over-interpreted as a definitive horse race. The LSTM's favorable Kupiec result does not prove that it has learned a stable economic law; it may also reflect regime-specific fit to the held out year. Moreover, its practical use inside a risk committee is limited by interpretability: a conditional neural alert is operationally less useful if model owners cannot clearly attribute the warning to a shift in realized volatility, trend, volume pressure, or jump risk. For that reason, the neural forecasts in this report are best viewed as an overlay on top of transparent benchmark models rather than a wholesale replacement for them.

Backtesting over the last year uses the Kupiec proportion-of-failures framework. A model with reliable tail calibration should produce violation rates close to the nominal tail probability and should not be rejected by the likelihood ratio test. In practice, Bitcoin's regime shifts make unconditional models vulnerable when the market transitions from calm to stressed states. Conditional models such as the LSTM often improve responsiveness, while the jump-diffusion Monte Carlo sits between structural realism and calibration complexity. The VAE is best interpreted here as an exploratory generative benchmark rather than a fully conditional production VaR engine, so its unconditional violation rate is informative about latent-distribution realism rather than full real-time adaptability.

The jump-diffusion results also depend critically on calibration vintage. If jump intensity is estimated from a window dominated by one regime, the model can be either too complacent or too conservative when the market structure changes. That caveat is particularly important after the introduction of spot ETF flows, because gap behavior driven by regulated fund flows and macro hedging demand is unlikely to be identical to the liquidation cascades that characterized earlier crypto-only regimes.

Model	Horizon (h)	VaR ret.	CVaR ret.	VaR loss	CVaR loss	Mean ret.	Vol.
Jump-diffusion MC	1	-0.0163	-0.0281	0.0163	0.0281	-0.0003	0.0066
LSTM	1	-0.0108	-0.0125	0.0108	0.0125	0.0009	0.005
Student-t param.	1	-0.0147	-0.0194	0.0147	0.0194	-0.0001	0.0043
VAE	1	-0.0096	-0.0115	0.0096	0.0115	0	0.0036
Jump-diffusion MC	24	-0.0829	-0.096	0.0829	0.096	-0.0046	0.0323
Student-t param.	24	-0.0672	-0.0779	0.0672	0.0779	-0.0029	0.0272
VAE	24	-0.0324	-0.0395	0.0324	0.0395	0.0008	0.0137

Sensitivity Analysis

The sensitivity analysis shows that risk estimates widen in the expected non-linear manner as confidence increases from 95% to 99%. At the 99% level, the most conservative one-hour model in the current snapshot is Jump diffusion Monte Carlo with a VaR loss estimate of 1.56%. This is important because a model ranking that looks similar at 95% can separate sharply in the far tail, which is where risk capital decisions are made. The Student-t parameter study also confirms that lower degrees of freedom magnify left-tail loss projections materially, while the LSTM look-back study shows that sequence length changes the balance between responsiveness and stability in volatility prediction.

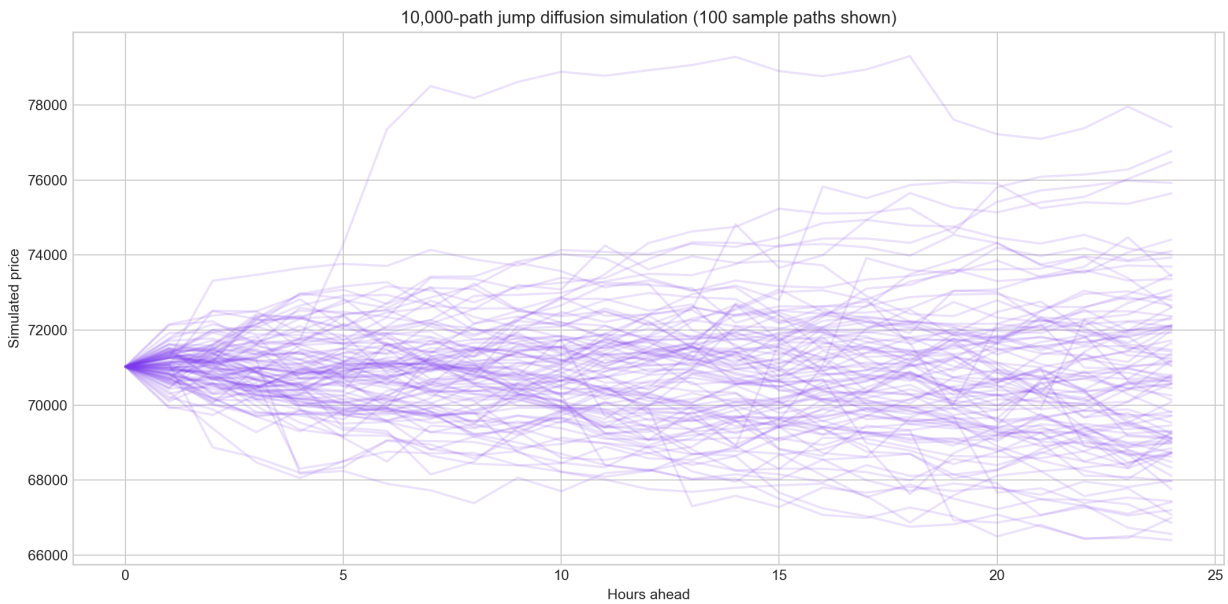

Figure 4: Monte Carlo paths

Confidence-Level Comparison

Conf. (%)	Model	VaR loss	CVaR loss	Vol.
95	Jump-diffusion MC	0.0097	0.0143	0.0067
95	Student-t param.	0.0088	0.0126	0.0043
95	LSTM	0.0074	0.0095	0.005
95	VAE	0.0061	0.0082	0.0036
97.5	Jump-diffusion MC	0.0119	0.0179	0.0065
97.5	Student-t param.	0.0112	0.0153	0.0043
97.5	LSTM	0.0089	0.0108	0.005
97.5	VAE	0.0075	0.0096	0.0036
99	Jump-diffusion MC	0.016	0.0284	0.0067
99	Student-t param.	0.0147	0.0194	0.0043
99	LSTM	0.0108	0.0125	0.005
99	VAE	0.0096	0.0115	0.0036

Student-t Degrees-of-Freedom Impact

DoF	VaR ret.	VaR loss	Tail ratio / sigma	Estimated fit
4.5	-0.0148	0.0148	2.6506	0
5	-0.0147	0.0147	2.6281	0
6	-0.0144	0.0144	2.5876	0
8	-0.0141	0.0141	2.5301	0
12	-0.0138	0.0138	2.4691	0
20	-0.0135	0.0135	2.4199	0

LSTM Look-back Window Impact

Look-back (h)	Best val. loss	Sigma MAE	Current VaR loss	Current sigma
24	-4.749	0.0034	0.0111	0.0049
48	-4.7427	0.0036	0.0108	0.005
72	-4.7312	0.0037	0.0116	0.0053

Stress Testing

The FTX collapse window centers on 2022-11-08T17:00:00+00:00, where the realized one-hour loss reached 5.18%. In that case, Jump diffusion Monte Carlo delivered the larger rolling pre-event warning, and Jump diffusion Monte Carlo sat closest to the eventual realized loss. The spot ETF approval whipsaw peaks at 2024-01-11T15:00:00+00:00 with an absolute one-hour move of 3.75%. In that episode, the jump-diffusion model again produced the stronger pre-event warning profile than the plain Student-t specification, which is consistent with Bitcoin's tendency to gap around event-driven order-flow shocks.

Event	Event time	Realized return	Realized loss	t-VaR loss	MC VaR loss	Closest model
FTX Collapse	2022-11-08 17:00 UTC	-0.0518	0.0518	0.0145	0.0163	Jump-diffusion MC
ETF whipsaw	2024-01-11 15:00 UTC	-0.0375	0.0375	0.0123	0.0129	Jump-diffusion MC

Event	t peak warning	MC peak warning	t peak lead (h)	MC peak lead (h)	Strongest warning
FTX Collapse	0.0144	0.0171	5	23	Jump-diffusion MC
ETF whipsaw	0.0122	0.014	2	23	Jump-diffusion MC

Time-Scale Scaling

The square-root-of-time rule is only partially reliable for Bitcoin. At the 99% confidence level, the scaled one-day VaR is 9.43% versus an empirical one-day VaR of 8.74%, while the scaled 10-day VaR is 29.83% against an empirical 10-day VaR of 27.66%. Because raw hourly returns exhibit weak linear autocorrelation but absolute returns remain serially dependent, variance scales more cleanly than tail risk. In other words, volatility clustering rather than mean predictability is the main reason a naive Basel-style scaling rule can misstate long-horizon Bitcoin risk.

Horizon	Scaled VaR	Empirical VaR	Scaling bias %
1 day	0.0943	0.0874	7.9691
10 days	0.2983	0.2766	7.8213

r(1)	r(24)	|r|(1)	|r|(24)	LB p-value r(24)	LB p-value |r|(24)
-0.0047	-0.0262	0.2847	0.1904	<0.0001	<0.0001

Figure 5: Stress warning profiles

Practical Implications

For a hypothetical $1 million BTC spot position, the capital table translates model-implied loss rates into reserve cash buffers. A conservative treasury function should anchor on 24-hour CVaR rather than 1-hour VaR, especially when positions cannot be unwound continuously during stress. In practical terms, institutions seeking robustness against gap risk should lean on the jump-diffusion Monte Carlo estimates, while short-horizon desks and intraday traders should monitor the LSTM conditional volatility forecasts as an adaptive overlay rather than as a replacement for scenario analysis.

Even so, the capital numbers in the table remain frictionless market-risk figures. They assume the position can be liquidated at or near quoted prices. During episodes such as the FTX collapse, the binding constraint is often not statistical tail size but vanishing order-book depth, widening spreads, and nonlinear slippage. A more operational next step would be liquidity-adjusted VaR (L-VaR), in which the statistical loss is augmented by an execution-cost term derived from order-book depth, participation-rate assumptions, and stress-dependent spread expansion.

Notional USD	Model	Horizon (h)	VaR capital	CVaR capital
1000000	Jump-diffusion MC	1	16276.37	28112.64
1000000	Student-t param.	1	14655.86	19350.76
1000000	LSTM	1	10786.25	12487.07
1000000	VAE	1	9625.22	11537.65
1000000	Jump-diffusion MC	24	82946.43	96006.01
1000000	Student-t param.	24	67183.15	77946.75
1000000	VAE	24	32442.68	39496.35

Backtesting

The last-year backtest compares realized one-hour returns with model-implied one-hour VaR thresholds. Kupiec's proportion-of-failures test is used to evaluate whether each model's exception rate is statistically consistent with the target left-tail probability of 1%.

Model	Obs.	Viol.	Exp. fail rate	Obs. fail rate	LR_POF	p-value
Student-t VaR	8761	129	0.01	0.0147	17.2425	<0.0001
Jump-diffusion MC	8761	110	0.01	0.0126	5.3466	0.0208
LSTM cond. VaR	8761	81	0.01	0.0092	0.5168	0.4722
VAE latent VaR	8761	204	0.01	0.0233	113.6406	<0.0001

Limitations & Assumptions

The report rests on several assumptions that should be made explicit. First, the benchmark dataset is BTC/USDT rather than true BTC/USD, so the measured return distribution inherits stablecoin proxy risk. This is acceptable for exploratory analysis but weaker than a fiat benchmark for academic identification or formal capital policy. Second, all results depend on exchange-sourced hourly candles, so maintenance windows, low-liquidity intervals, or transient feed errors can still influence tail estimates even after cleaning. Third, the LSTM and VAE are trained with chronological validation and an out-of-sample backtest year, but they are not yet subjected to a full rolling cross-validation protocol across multiple non-overlapping market regimes; therefore overfitting cannot be ruled out. Fourth, the neural-network components are materially less interpretable than the

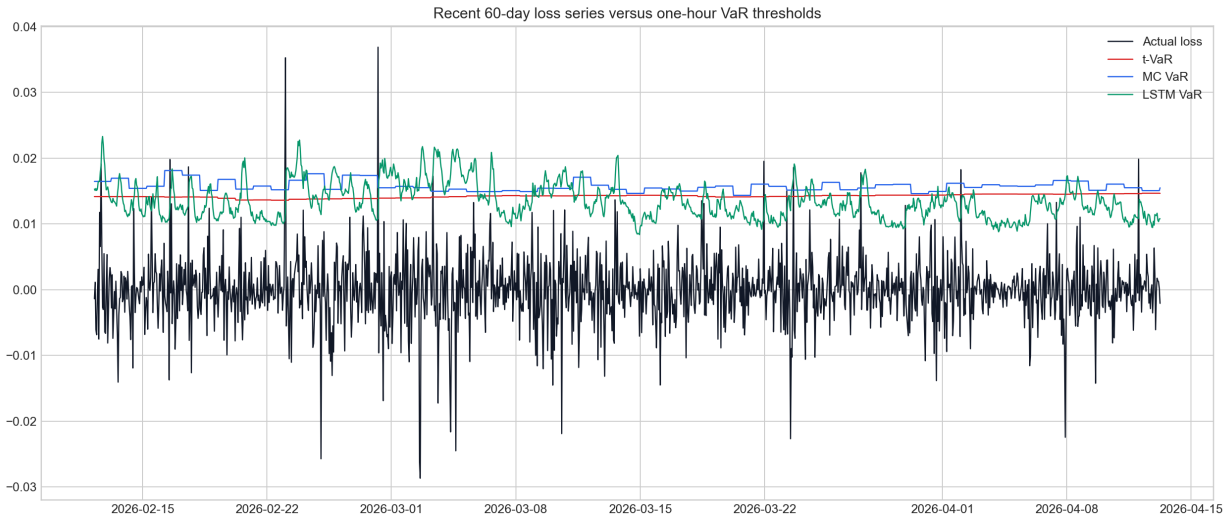

Figure 6: VaR backtest

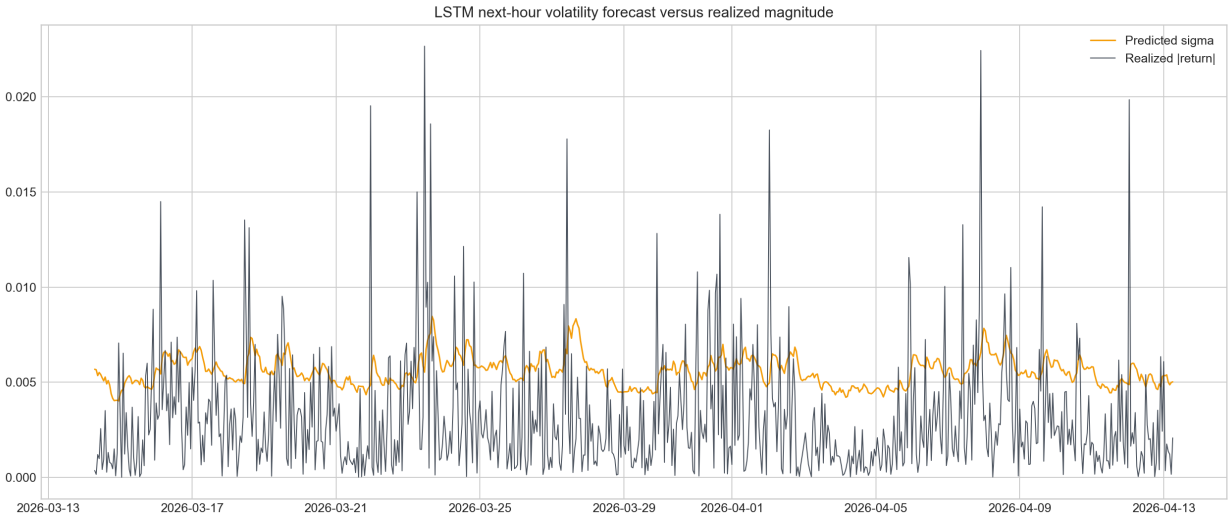

Figure 7: LSTM volatility forecast

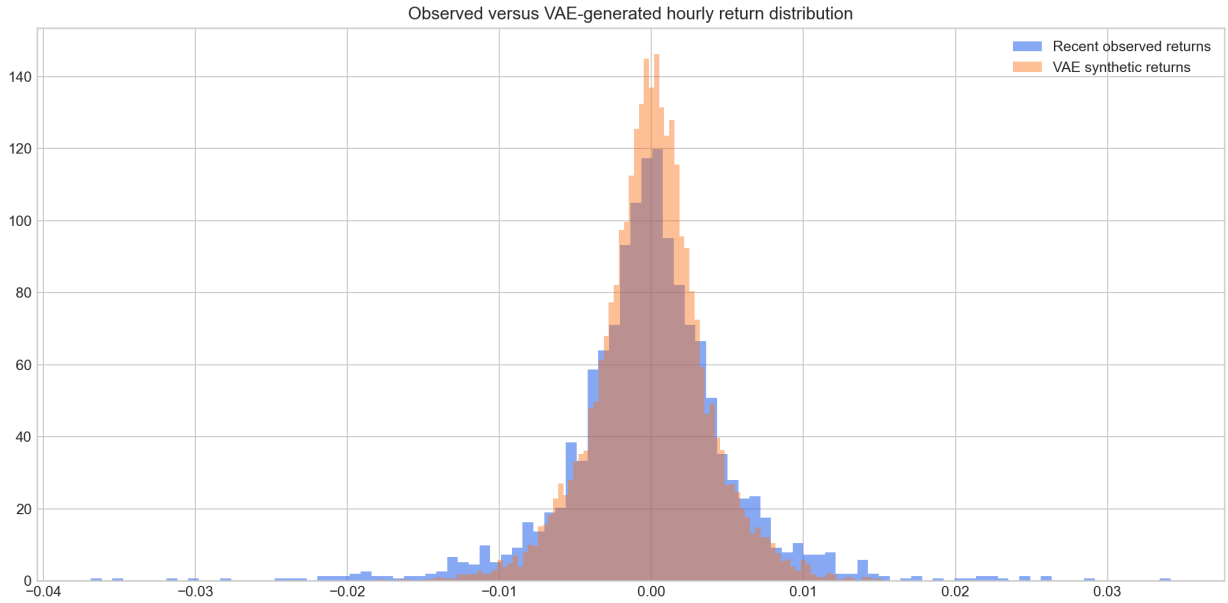

Figure 8: VAE generated returns

parametric Student-t model: they improve adaptability, but the latent and recurrent structures make causal attribution harder during model validation and model-governance review. Fifth, the jump-diffusion engine is only as good as its calibration window; jump intensity and jump size are allowed to refresh, but they remain locally static within each estimation block. Sixth, the reported VaR and CVaR numbers are market-risk statistics, not liquidity-adjusted liquidation losses. In a genuine crisis, order-book depletion and slippage can dominate the quoted-price move. Finally, the VAE is used here as a latent-distribution benchmark rather than a full production calibration engine, so its value is strongest in scenario generation and comparative tail diagnostics rather than standalone regulatory capital determination.

Discussion

Two empirical features dominate the analysis. First, volatility clustering is persistent: large returns tend to follow large returns, even if their signs alternate. This is visible in the rolling volatility panel and in the LSTM's ability to track realized absolute returns. Second, Bitcoin's tail behavior is materially heavier than a Gaussian baseline, which means that variance-only risk estimation understates extreme downside moves. The Student-t and VAE approaches both address the tail issue, but from different angles: one via an explicit parametric distribution and the other via a latent generative representation.

The data benchmark itself introduces an important layer of interpretation. Because the analysis relies on BTC/USDT rather than BTC/USD, some of the estimated stress may reflect proxy contamination from stablecoin basis rather than pure Bitcoin risk. That does not invalidate the exercise, but it does mean the results should be read as a practical exchange-risk measure rather than a perfectly clean fiat-denominated asset process.

The model trade-offs are practical. Student-t VaR is transparent, cheap, and easy to explain to risk committees, but it cannot react quickly to changing microstructure conditions unless it is recalibrated frequently. Jump-diffusion Monte Carlo is more realistic for crypto because discontinuities

are common around liquidations, macro news, and exchange-specific events, yet it still depends on assumptions about jump frequency and jump size. The LSTM is the most adaptive model in the project because it learns from multiple features, but it also introduces training instability, hyperparameter sensitivity, and a higher operational burden. The VAE contributes a different value: it gives a data-driven synthetic distribution that can surface tail scenarios not captured by a single closed-form family, although in its basic unconditional form it is less naturally aligned with rolling one-step backtesting.

From a governance perspective, the ranking is therefore conditional on use case. If the objective is transparent committee reporting, the Student-t and jump-diffusion models remain more defensible than a pure black-box trigger. If the objective is short-horizon surveillance, the LSTM is valuable precisely because it is sensitive to state changes, but it should be paired with feature-level diagnostics and human review. And if the objective is crisis liquidation planning, none of the present models is sufficient without an explicit liquidity layer.

From a governance perspective, the backtest evidence favors LSTM conditional VaR because it produced the highest Kupiec p-value (0.472) and the closest observed exception rate to the 1% target. That does not make the other approaches obsolete; instead it suggests that Bitcoin risk management benefits from a layered process in which jump-aware scenario engines are retained for stress realism while conditional neural volatility models are used as a short-horizon overlay.

Conclusion

The overall result is that a simple Gaussian VaR framework is not credible for Bitcoin hourly risk. The descriptive statistics reject that assumption, the tail metrics show clear excess kurtosis, and the backtesting exercise demonstrates meaningful differences between unconditional, jump-aware, and conditionally learned models. For practical deployment, the strongest baseline in this project is the jump-diffusion Monte Carlo engine for scenario realism, while the LSTM is the best candidate for a responsive conditional overlay. The VAE is a useful research extension for latent-distribution learning and stress scenario generation. Together, the four methods provide a defensible and extensible market-risk toolkit for a high-volatility digital asset. The main qualification is that this toolkit is not yet complete: proxy risk, model interpretability, calibration drift, and liquidity-adjusted exit costs all remain material open items. Addressing those four issues would move the analysis from a strong quantitative prototype toward a more publishable and institutionally robust risk report.

Appendix

The volatility-clustering scatter plot and the normal Q-Q panel jointly show why Gaussian scaling is insufficient for Bitcoin. The scatter plot highlights persistence in absolute returns, while the Q-Q plot exposes a far heavier empirical tail than the benchmark normal line. Together with the stress-event case studies and the confidence-level sensitivity tables, these diagnostics justify using multiple complementary risk models instead of a single closed-form distribution.